

Enterprise HRMS Training Suite

Final Internship Project Documentation

Developer: Sanidhya Vijayvargiya

Repository: <https://github.com/sanidhya254/sanidhya-demo>

Project Overview

Enterprise HRMS is a full-stack web application built to simulate real HRMS and ERP modules used by companies. It includes authentication, employee management, leave workflow, asset tracking, dashboard analytics, reports, notifications, and audit logs. The project is designed for final internship delivery and deployment on GitHub, Render, Vercel, and Neon PostgreSQL.

Objectives

- Build a complete React and Node.js full-stack application.
- Apply JWT authentication and protected API routes.
- Demonstrate employee CRUD, SQL relationship design, dashboard graphs, and reports.
- Prepare deployment-ready project artifacts for GitHub, Render, Vercel, and PostgreSQL.
- Create final documentation and presentation for interview/project submission.

Technology Stack

Layer	Technology
Frontend	React, Vite, Recharts, Lucide React
Backend	Node.js, Express.js
Authentication	JWT, bcryptjs
Uploads	Multer
Database Design	PostgreSQL schema with relationships, JOINS, JSONB audit logs
Deployment	GitHub, Render, Vercel, Neon PostgreSQL

Application Modules

- Authentication: login, signup, hashed passwords, JWT token protection.
- Dashboard: total employees, departments, uploaded files, leave requests, asset count, salary and distribution charts.
- Employee Management: create profiles, assign departments, skills, salary, city, domain, work mode, and attendance.
- Leave Workflow: manager and HR approval/rejection with final status tracking.

- Asset Management: laptop, monitor, ID card, access card allocation and status tracking.
- Reports: salary, TDS, PF, net salary, attendance split, and CSV export.
- Student Filters: semester, city, domain, online/offline/hybrid mode.
- Audit Trail and Notifications: enterprise-style records for data changes and system events.

Database Design Summary

Table	Purpose
users	Stores login users and roles.
departments	Department master records.
skills	Skills master records.
employee_profiles	Employee HR profile details.
employee_images	Multiple document/image uploads.
employee_skills	Many-to-many mapping between employees and skills.
leave_types / leave_balance	Leave master and employee balances.
leave_applications / approval_history	Workflow and approval records.
assets / asset_allocations	Asset inventory and allocation lifecycle.
notifications	System notification engine.
audit_logs	JSONB old/new value audit trail.
students	Training/student filter data.

API Endpoints

Method	Endpoint	Use
GET	/api/health	Backend health check.
POST	/api/auth/login	User login and JWT generation.
POST	/api/auth/signup	Create new user.
GET	/api/bootstrap	Protected dashboard and module data.
POST	/api/employees	Create employee profile.
PUT	/api/employees/:id	Update employee profile.
DELETE	/api/employees/:id	Delete employee profile.
POST	/api/employees/upload	Upload up to 5 files.
POST	/api/leaves/:id/action	Approve or reject leave.
GET	/api/reports/salary.csv	Export salary report.

Deployment Plan

- Push final source code to GitHub repository sanidhya254/sanidhya-demo.
- Create Neon PostgreSQL database and run DATABASE_SCHEMA.sql.
- Deploy backend on Render using build command npm install and start command npm start.
- Set Render environment variables: JWT_SECRET, DATABASE_URL, FRONTEND_URL, NODE_ENV.
- Deploy frontend on Vercel with VITE_API_URL set to the Render API URL.
- Test complete application: login, employee module, leave workflow, assets, reports, filters, and API health.

Final Submission Links

Item	Value
GitHub Repository	https://github.com/sanidhya254/sanidhya-demo
Live Frontend URL	Pending Vercel deployment
Live Backend API URL	Pending Render deployment
Database Schema	DATABASE_SCHEMA.sql
Documentation PDF	docs/PROJECT_DOCUMENTATION.pdf
Presentation PPT	docs/Enterprise_HRMS_Presentation.pptx
Screenshots	docs/screenshots

Screenshots

Dashboard

Enterprise HRMS

Workflow Engine

Dashboard

Employees

Leave Workflow

Assets

Reports

Students

Audit

Logout

ROLE: ADMIN

Dashboard

Filter semester, city, domain, attendance...

Employees

6

Allocated Assets

2

Pending Leaves

2

Students

5

Employees

Enterprise HRMS

Workflow Engine

Dashboard

Employees

Leave Workflow

Assets

Reports

Students

Audit

Logout

ROLE: ADMIN

Employees

Filter semester, city, domain, attendance...

Create Employee Profile

name

phone

city

designation

salary

Full Stack

Software Development

Present

Choose files

No file chosen

+ Save Employee

Employee Listing with JOIN Data

NAME	DEPARTMENT	SKILLS	SALARY	STATUS
Admin User Director	Software Development	React, NodeJS, PostgreSQL	₹1,50,000	Present

Leave Workflow

Enterprise HRMS
Workflow Engine

Dashboard

Employees

Leave Workflow

Assets

Reports

Students

Audit

Logout

ROLE: ADMIN

Leave Workflow

Filter semester, city, domain, attendance...

Manager Approval, HR Final Approval & Leave Balance

Amit Patel

Casual Leave • 2 day(s)

2026-06-12 to 2026-06-13

Approved

Pending

Pending

Approve

Reject

Neha Jain

Sick Leave • 1 day(s)

2026-06-10 to 2026-06-10

Approved

Approved

Approved

Approve

Reject

Rohit Singh

Assets

Enterprise HRMS
Workflow Engine

Dashboard

Employees

Leave Workflow

Assets

Reports

Students

Audit

Logout

ROLE: ADMIN

Assets

Filter semester, city, domain, attendance...

Asset Management System

LAP-1001

MacBook Air M4

Laptop • Rahul Sharma

Allocated

MON-2042

Dell 24 Monitor

Monitor • Amit Patel

Allocated

IDC-9001

Office ID Card

ID Card • Neha Jain

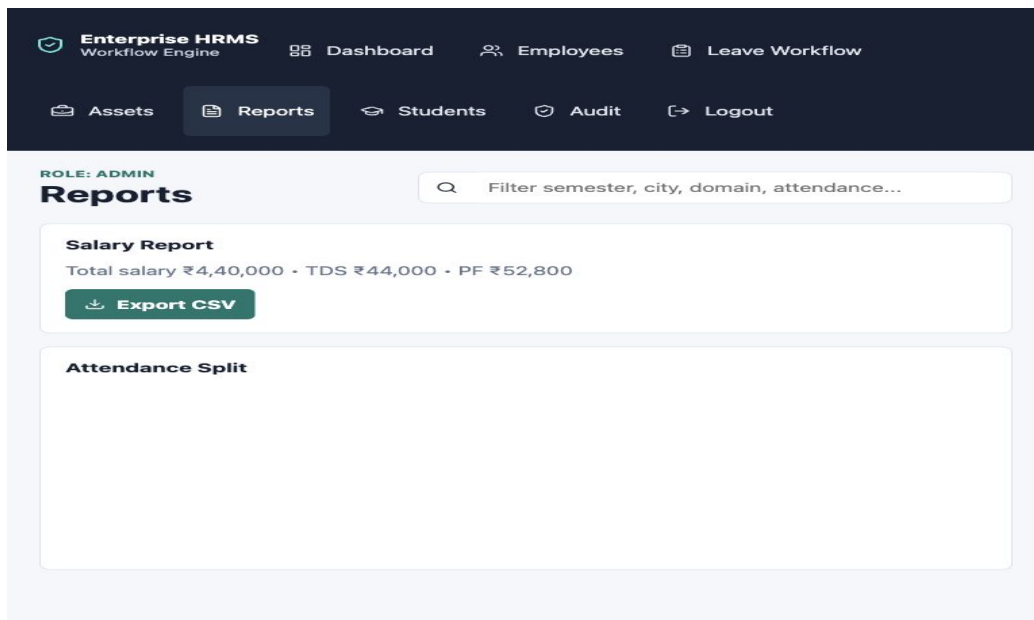
Returned

KEY-5010

Access Card

Access • Unassigned

Reports



Future Enhancements

- Connect all seed data to live PostgreSQL repositories and migrations.
- Add role-specific screens for Admin, HR, Manager, and Employee.
- Add email notifications and cloud file storage.
- Add Swagger/OpenAPI documentation page.
- Add automated tests and CI/CD pipeline.